

DATA ANALYSIS USING SQL

QUERY, EXPLORE AND UNDERSTAND YOUR DATA

Structure it. Query it. Explore it — with AI as your assistant.

Intermediate Graduate



WHO IS THIS FOR?

ASPIRING DATA & BI ANALYSTS

Students aspiring to become Data Analysts or BI Professionals, starting from the database layer.

DATA-CURIOUS PROFESSIONALS

Professionals looking to upskill in data visualization, and anyone who wants to improve productivity and decision-making using data.

COURSE MODULES



Generative AI Basics

- LLMs, NLP, prompting
- AI as a data analyst / SQL expert



Introduction to SQL

- SQL fundamentals & entities
- Data retrieval & joins



EDA with AI

- Data profiling & cleaning
- Pattern & outlier detection



Excel as a Data Source

- Import query results
- Analyze in Excel

CORE SKILLS YOU WILL PRACTICE



Query Writing



Table Joins



EDA with AI



Data Import



Insight Reporting

TOOLS & TECHNIQUES YOU WILL USE

- SQL** **SQL Fundamentals**
Core querying language for relational data
- Jn** **Joins & Data Retrieval**
Combine tables, filter results
- AI** **Generative AI Prompting**
AI as your SQL assistant
- EDA** **AI-Assisted EDA**
Profiling, cleaning, pattern discovery
- Ex** **Microsoft Excel**
Import & analyze query results
- DB** **Database Design Basics**
Tables, keys & relationships
- Qo** **Query Optimization**
Writing efficient, readable SQL

DURATION

16

HOURS

Total instructor-led training time

Instructor-led · Outcome-based learning · Success factor

KEY OUTCOMES

- QUERYING**
Confidently write and run SQL queries
- ANALYSIS**
Explore data and uncover patterns with AI
- COMMUNICATION**
Present findings clearly to stakeholders

WHAT YOU WILL LEARN

- ✓ Understand database structure and relationships
- ✓ Write SQL queries, retrieve data, and use joins
- ✓ Use AI tools to accelerate exploratory data analysis
- ✓ Detect patterns, outliers and trends in data
- ✓ Bring query results into Excel for further analysis

YOU WILL WALK AWAY WITH

- ✓ Practical skills you can apply immediately
- ✓ Ready-to-use tool & technique knowledge
- ✓ Improved productivity & efficiency
- ✓ Confidence querying and exploring databases
- ✓ A completion certificate for your resume



HANDS-ON TRAINING

Real-world exercises and practical use cases



EXPERT-LED SESSIONS

Learn from experienced practitioners



RESOURCES & TEMPLATES

Curated guides, templates & practice files



CERTIFICATE OF COMPLETION

Showcase your new skills